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Enhancing Predictive Maintenance for Heavy Vehicles: A Multi-Stage Feature Selection and Preprocessing Framework on the Scania Dataset

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Abstract

In the newer vision of Industry 5.0, the industrial transport industry is evolving towards paradigms that prioritise the adoption of intelligent predictive maintenance systems. These systems enable the prediction of assets at risk of imminent failure, thus allowing maintenance to be planned before failure. This study utilises a real-world dataset provided by Scania to predict the imminent failure of an anonymised engine component in heavy-duty trucks. The dataset presents common real-world challenges, including anonymised values, class imbalance, and missing values. A multi-stage feature selection and preprocessing framework is proposed to address these challenges. A comparative analysis of several machine learning models is conducted to evaluate predictive performance. The findings indicate that the framework improves model robustness and transparency. Moreover, by enabling more accurate failure prediction, the approach contributes to a tangible reduction in maintenance and/or replacement costs associated with the monitored component.

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Keywords: Industrial challenge, SCANIA Component X, Predictive Maintenance

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1. Introduction

Modern transportation systems are equipped with sensors that can continuously collect and analyse data on driving behaviour and vehicle performance. The integration of emergent technologies, such as Big Data, Machine Learning (ML), and Internet of Things (IoT), within the automotive industry has facilitated real-time monitoring and predictive analytics for automotive diagnostics [1]. Data acquisition may include engine temperature, speed profiles, maintenance logs, and event-based records, including braking behaviour and safety system activations. By analysing these data, predictive systems can be developed to anticipate component failures and support the planning of maintenance interventions aimed at preventing unexpected breakdowns. The identification of potential failure plays an important role in ensuring the safety and reliability of a vehicle on the road [2]. It can reduce downtime, minimize maintenance costs, and extend the lifespan of components, thereby promoting more sustainable vehicle operation through reduced resource consumption and waste generation. The adoption of an ML prediction model for planning maintenance operations has been demonstrated to result in a reduction in financial expenditure for automotive production [3]. The integration of ML and IoT in the context of warranty management has the potential to optimise resources, enhance customer experience, and maintain high levels of competitiveness for automotive companies [4].

Despite the great potential of ML techniques for constructing accurate predictive models, significant challenges exist in the automotive sector. The high dimensionality and heterogeneity of vehicle data present significant challenges to data integration and preprocessing [5]. The data may originate from different sources, including onboard sensors, telematics systems, and maintenance records. However, due to concerns regarding privacy and competitiveness, companies are often reticent to share data [6], thus resulting in a lack of available datasets and, consequently, a scarcity of data for the development of a robust predictive maintenance model. Even when data are accessible, a major challenge arises from the intrinsic imbalance between normal operating conditions and failure events. Modern vehicles are designed to be highly reliable, making critical component failures infrequent [7]. Moreover, when failures occur, they are not always recorded in sufficient detail, creating further gaps in historical failure data. This imbalance can significantly affect the performance of predictive models, which typically require a balanced representation of both faulty and non-faulty instances for effective training. As a result, the lack of failure examples can lead to biased models with reduced predictive accuracy.

To address the challenges of automotive data, a robust framework is required to define an accurate predictive model. One of the main ML frameworks noted in the literature is the Knowledge Discovery in Database (KDD) [8]. This framework is developed through a series of well-defined steps ranging from the collection and preparation of data to the interpretation of the results obtained. Five main phases can be identified from the framework, as shown in Fig. 1.

The first phase consists of the selection and integration of data coming from different sources. In the second phase, the preprocessing step is computed to handle missing values, duplicates, and outliers. The next phase consists of the conversion of data into an appropriate form for the analysis to be conducted. Conversion refers to normalization, standardization, feature selection, and feature engineering. The objective is to define an informative dataset to be used as input data for the ML model in the next phase. The core phase of the KDD involves training and testing ML techniques to develop accurate predictive models for the problem under investigation. Finally, the interpretation of results is necessary to identify the most accurate model. Although these phases are generally executed sequentially, there is frequently a necessity to repeat prior steps to enhance the accuracy of the predictive model.

This paper focuses on addressing the challenges of a real-world dataset for predicting failures of a component in truck engines manufactured by SCANIA AB in Sweden [9]. SCANIA AB is a leading global manufacturer of heavy-duty trucks, widely recognized for its technological innovations and large-scale production of commercial vehicles. A multi-stage framework for the preparation of the dataset and the training and validation of a predictive model is adopted. A comparative analysis of several machine learning models is conducted to evaluate predictive performance. Results demonstrated superior performance in comparison to previous studies in the same dataset, thereby demonstrating the robustness of the adopted framework.

The rest of the paper is organized as follows: Section 2 provides details about the challenge proposed by SCANIA AB for the prediction of failures of a component in truck engines; Section 3 illustrates the methodology adopted to address the challenge; Section 4 presents the results and discussion, comparing the obtained outcomes with those reported in previous studies that addressed the same SCANIA AB challenge, highlighting similarities, differences, and potential improvements. Finally, Section 5 provides the conclusion of the work, summarizing the key insights and outlining possible directions for future work.

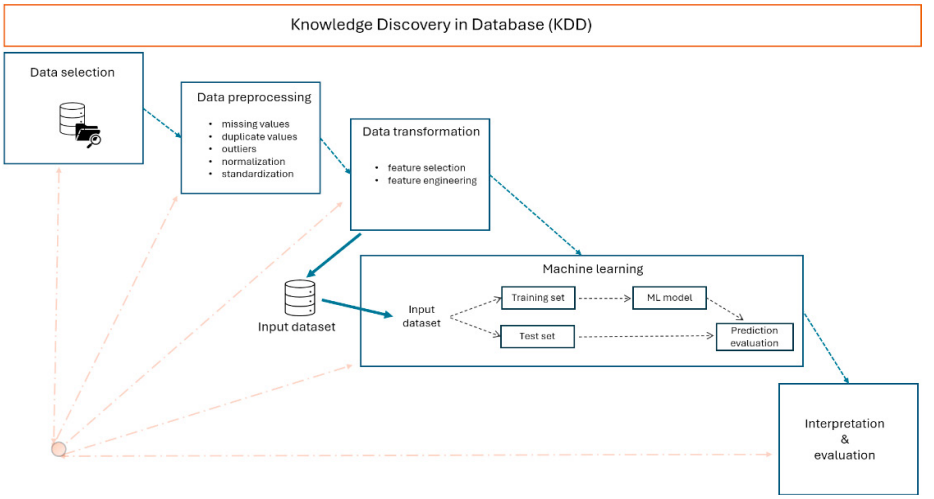


Fig. 1 Main phases of Knowledge Discovery in Database framework.

2. Problem definition

In this work, a predictive maintenance problem is addressed. The problem involves predicting a failure event for an engine component of heavy-duty trucks by analyzing historical data on operational efficiency and repair records of the component. This task was proposed as part of the Intelligent Data Analysis Industrial Challenge 2024, where a real-world dataset was provided by SCANIA AB [9]. The objective of the challenge was to develop an accurate predictive model for failure events, classified into five categories according to their remaining time until failure. These categories are detailed in Table 1.

Table 1 Categories of observations based on their remaining time until a failure event.

Classes	Time until failure
0	No failure event
1	within 48 to 24 time units
2	within 24 to 12 time units
3	within 12 to 6 time units
4	within 6 to 0 time units

The performance of the predictive models are evaluated in terms of financial expenditure for the company, considering the impact of unnecessary checks made by a mechanic during a truck inspection, or the consequences of late identification of component failure, which may lead to a breakdown and an expensive rescheduling of the transport to which the truck was assigned. The financial expenditure is computed by using a cost matrix of miss-classification, as demonstrated in Table 2, multiplied by the number of instances classified correctly or incorrectly by the predictive model. An asymmetric cost behavior is highlighted between false negatives and false positives. A false negative is a prediction of failure as no failure, while a false positive is a prediction of no failure as failure. In detail, higher costs are assigned to false negatives, indicating that the rescheduling of transport following a breakdown is more expensive than unnecessary maintenance interventions. This cost-sensitive evaluation framework reflects real-world industrial priorities, where avoiding unexpected failures is critical to maintaining operational efficiency, reducing downtime, and improving customer satisfaction.

2.1. *Dataset*

The real-world dataset provided by SCANIA AB is considered in this study. Its release aimed to address the lack of real-world benchmark datasets in the literature. The proposed problem to be solved is to predict the risk of failure

for an engine component (called Component X) in heavy-duty trucks. The dataset includes operational data (i.e., sensors installed on trucks to monitor the behavior of Component X), truck specifications (such as engine type and wheel configuration), and repair records. All the data were anonymized and selected by experts. The complete dataset was divided into three subsets, each containing the three distinct data types:

1- Operational data: It contains 107 variables. The first variable ('*variable_id*') identifies the specific truck and can be used as the key variable to extract the necessary information from the remaining two subsets. The second variable ('*time_step*') quantifies the number of time units the vehicle and Component X have been in use. The remaining anonymized variables are organized into a series of numerical counters (with a total of eight variables) and histogram bins (six variables with 10, 10, 11, 10, 20, and 36 bins, respectively). Across time steps, the values in each bin are cumulative values representative of the frequency of the registered values in that bin;
2- Specifications: It contains the key variable '*variable_id*' and eight categorical variables, where each category is anonymized as Cat0, Cat1,…, Cat9. A total of 3607 unique combinations of these variables are provided, identifying unique truck models;
3- Repair records: It contains three variables. The first is the key variable '*variable_id*'. Then, the variable '*length_of_study_time_step*' is a numerical variable representative of time units of operation. The last is a binary variable: a value of 0 represents no repair intervention during the '*length_of_study_time_step*'; a value of 1 represents a repair record after '*length_of_study_time_step*' time units.

Table 2 Cost matrix for miss-classification.

Actual/Predicted	Class 0	Class 1	Class 2	Class 3	Class 4
Class 0		7	8	9	10
Class 1	200		7	8	9
Class 2	300	200		7	8
Class 3	400	300	200		7
Class 4	500	400	300	200	

These three subsets were then further subdivided into the three sets of training, validation, and test. Table 3 contains information about the dimensions of these sets.

Table 3 Main information about training, validation and test sets.

	Training set	Validation set	Test set
# vehicles	23 550	5046	5045
# observations	1 122 452	196 227	198 140
# mean observations per vehicle	48	46	45

For validation and test sets, the binary variable in the repair records subset is substituted with five classes, created by a time window before the failure event of Component X occurs, as described in Table 1.

As a real-world scenario, the dataset presents different challenges:

1- Operational data were acquired by appropriate sensors installed on the trucks that collect real-time data and store them in the vehicle control units. Sometimes, losing connection with devices that collect data can provide some errors (such as a reset of the numerical counter). A post-processing phase of the data was performed by covering the majority of the possible types of corrupted data. However, a systematic analysis revealed that 282,425 corrupted data remain in the training set, impacting 25,432 observations across the dataset;
2- Missing values are presented in the training set with a total of 354,634 missing values in 22,481 observations of the training set (2% of the total number of observations);
3- An imbalanced distribution between registered failures and non-failure components emerged. As shown in Fig. 2a, 90.4% of the observations are associated with non-failure components while the remaining 9.6% is divided into the 4 categories of failure, as shown in Fig. 2b.

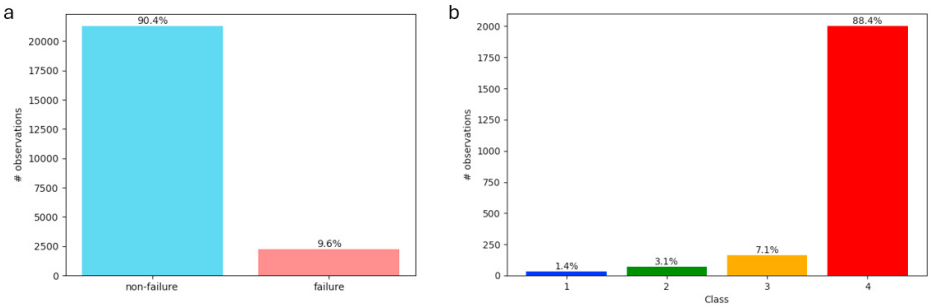


Fig. 2 Target variable distribution between a) failure and non-failure observations and b) the four possible failure types.

3. Methodology

With the objective of developing a predictive model able to anticipate failure events for Component X, a structured framework is necessary to address the challenges of a real-world dataset. Following the steps outlined in the KDD, the framework ensures robust data preparation, feature selection, and model development. The framework adopted in this study consists of five main phases and 10 steps, as illustrated in Fig. 3. During its implementation, a backward-forward process is computed several times to improve the performance of the predictive models developed.

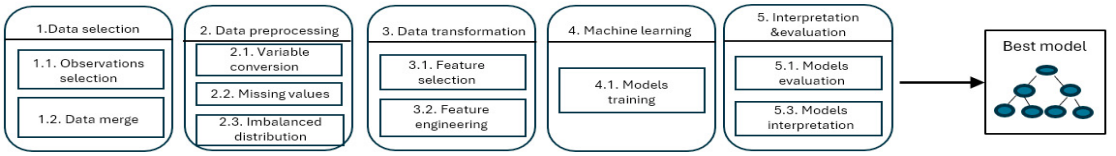


Fig. 3 Framework adopted.

3.1. Data selection

The first phase of the adopted framework involves the two steps of data selection and integration. It consists of the selection of observations and their subsequent merging to form a unified dataset. All relevant information must be selected in this phase concerning the problem to solve. Concerning the SCANIA challenge addressed in this study, each monitored vehicle is characterized by several time steps in which the upload of the data stored in the vehicle control units is made. The monitoring of the vehicle is terminated either when a repair intervention is required or after a specified number of time steps have elapsed without a maintenance record being recorded. To predict when Component X is approaching failure, given the highly imbalanced distribution between no failure and failure events, it was necessary to select only the last observation for each vehicle from the operational data. Thereafter, a merge of the three subsets (i.e., operational, specifications, and repair records) was made by the key variable *'vehicle_id'*, as illustrated in Fig. 4. The merged data for the training set consists of 23,550 observations and 115 variables at the end of this phase.

3.2. Data preprocessing

To ensure that all the relevant data are retained for constructing accurate predictive models, a data preprocessing phase is necessary. It includes the handling of missing values, the correction of sensor measurements due to lost connections, the standardization of numerical variables, and the handling of the imbalanced distribution issue.

In detail, the following preprocessing tasks were performed in this study:

1- The categorical variables related to the specifications of the vehicle were combined into a single variable (*"model"*) representative of the model of the vehicle. 3,607 unique models were identified in the training set. This allowed the analysis of similar behavior between vehicles of the same model;

2- A total of 21 missing values emerged from the merged training set. Two different strategies were adopted for their handling. If the missing value relates to a numeric variable in the operational data, for which two previous values exist, then the missing value is replaced according to the following formula:

$$value_i = \frac{value_{i-1}+value_{i-2}}{2} + (value_{i-1} - value_{i-2}) \quad (1)$$

If the missing value relates to a numeric variable in the operational data and no two previous values exist, a regression model is trained on the vehicles of the same model to replace the missing value with a prediction. These imputation strategies were selected to preserve the temporal dynamics of the signals and to exploit similarities across vehicles of the same model when local information was insufficient. Although such methods may inherently introduce some bias, the limited number of missing values and the consistency of results obtained with alternative imputations indicate that their impact on the overall analysis is negligible. To train the model, the mixed linear model of the statsmodels package [10] of Python was used. To improve the convergence capability of the adopted ML models, a standardization of the numerical variables was computed;

3- The problem under study presents a significant challenge of imbalanced distribution, which can drastically affect the predictive performance of ML models. To address this challenge, the SMOTE technique [11] was adopted as a sampling method to balance the dataset.

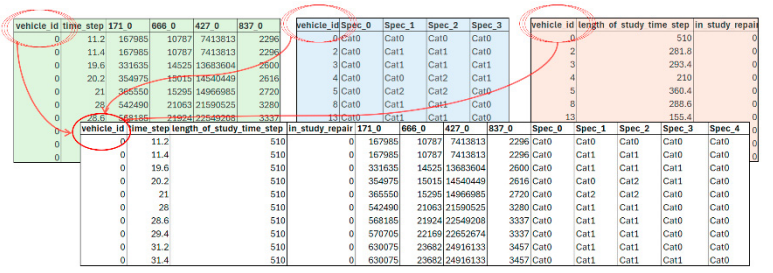


Fig. 4 Merging operation of the three provided subsets.

3.3. Data Transformation

The feature selection process has the main goal of identifying the most informative variables for prediction, reducing dimensionality and preventing overfitting. In this study, all the numerical variables of the operational data were selected, while the categorical variables of the specifications data were removed. Then, a feature selection technique was adopted to reduce dimensionality. A model-based feature selection technique is used in this study. We trained a Random Forest [12] with 100 estimators.

Finally, a feature engineering task was performed for the correct definition of the target variables. The training set is characterized by a binary target value, while the validation and test sets contain a multiclass target variable. To align the three datasets, the binary variable was converted into a multiclass one, according to the remaining time until failure, as described in Table 1. This process identified ten new missing values, given by the presence of observations with failure but after a period of more than 48 hours. Given the imbalanced distribution of the data and the higher cost associated with false negatives, we decided to keep these ten observations by assigning them the label that identifies a failure event within 24 and 48 hours. A total of 53 variables were obtained at the end of the data transformation phase. Fig. 5 shows the selected variables concerning the importance (i.e., how much a variable contributes to improving the model's predictions).

3.4. Machine Learning

Given the informative dataset obtained by performing the first three main phases of the KDD, the next step involved training predictive models to forecast failure events of Component X accurately. In this study, four different ML models were trained and tested: Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Support Vector Machine (SVM).

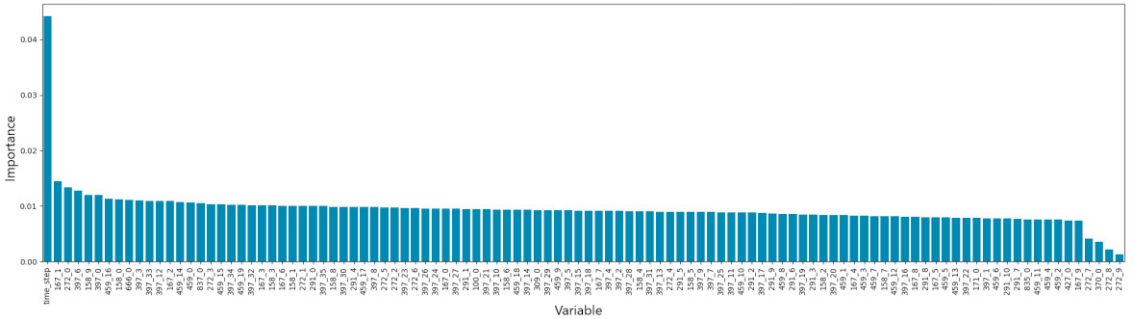


Fig. 5 Selected variables based on their importance.

The Decision Tree model is a tree-based approach in which partitioning rules are defined based on the values of the input variables. Unlike many other popular ML models, DT is considered a white-box model, as it offers transparency in its decision-making process and enables interpretability of the rules used for class prediction [13]. In contrast, DT may suffer from overfitting, particularly when applied to complex and high-dimensional data. Ensemble approaches, such as Random Forests and Boosting, are commonly employed to avoid the limitations of an individual model. These approaches combine multiple weak learners into a more robust predictive model capable of capturing complex relationships within the data more efficiently. Tree ensemble models demonstrated good prediction performance on tabular data. However, they reduce transparency and interpretability [14]. SVM is one of the most powerful and robust classification and regression algorithms in multiple fields of application [15]. It is a kernel-based ML model. In this model, a hyperplane is identified to separate the classes by maximizing the margin (i.e., the distance between the hyperplane and the nearest data points from each class).

ML models are characterized by hyperparameters, which are external parameters not directly learned from the data but significantly influence the model's performance. To achieve optimal predictive performance, the setting of these parameters becomes crucial. In the literature, different hyperparameter optimization methods exist. One of the most powerful optimization frameworks for hyperparameter tuning is Optuna [16], which combines efficient searching and pruning algorithms. In this study, the Optuna framework is implemented to optimize the hyperparameter selection.

3.5. Model evaluation

To assess the effectiveness of the trained model, the employment of performance metrics is essential. The challenge addressed in this study proposes to consider a cost matrix to evaluate the performance of the ML models. This matrix should be multiplied by the confusion matrix that provides the number of true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) predictions of the model. The confusion matrix enables an understanding of the type of errors made by the model. In conjunction with the cost matrix, a final performance metric that weights the more important class for the problem under investigation is obtained. To obtain a more comprehensive understanding of the model performance, other key metrics can be computed from the confusion matrix, such as accuracy, precision and recall.

Accuracy provides the overall proportion of correct predictions. In the case of class imbalance, it can be misleading, given the capacity of the model to correctly predict the majority class but make many errors on the minority class, on which there are few observations. Precision measures the proportion of correctly predicted positive instances among all instances predicted as positive, while recall indicates the proportion of actual positive observations correctly identified by the model. The formulations for these three metrics are provided in Table 4. A combined analysis of these two metrics allows a balanced evaluation of the predictive capabilities of the ML models.

Table 4 Metrics formulation.

Accuracy	Precision	Recall
$acc = \frac{TP + TN}{TP + FP + TN + FN}$	$P = \frac{TP}{TP + FP}$	$R = \frac{TP}{TP + FN}$

4. Results

The problem under investigation was addressed as both a multiclass classification problem and a binary problem, distinguishing between faulty and non-faulty. The second approach was tested concerning the fact that a higher cost is associated with false positives. After preprocessing the dataset as described in Section 3, the training set was used to train the four ML models, while the validation set was used to compute the performance metrics and choose the best model for the problem under investigation. Additionally, the results on the validation set are compared with other state-of-the-art results for the problem under investigation. Finally, the prediction on the test set is provided. The adoption of the Optuna frameworks provided the best hyperparameters for each of the models trained, divided by the multi-class and binary approaches, as shown in Table 5.

Table 5 Hyperparameters selected for ML models.

ML model	Binary approach	Multi-class approach
Decision Tree	• max_depth: 6 • min_samples_split: 11 • min_samples_leaf: 8 • criterion: log_loss	• max_depth: 30 • min_samples_split: 15 • min_samples_leaf: 6 • criterion: Gini
Random Forest	• n_estimators: 263 • max_depth: 25 • min_samples_split: 13 • min_samples_leaf: 9 • max_features: sqrt • bootstrap: True	• n_estimators: 80 • max_depth: 25 • min_samples_split: 9 • min_samples_leaf: 10 • max_features: sqrt • bootstrap: True
XGBoost	• n_estimators: 84 • max_depth: 6 • learning_rate: 0.0674 • subsample: 0.5848 • colsample_bytree: 0.8013 • gamma: 0.0105 • reg_alpha: 0.0741 • reg_lambda: 0.8036	• n_estimators: 132 • max_depth: 11 • learning_rate: 0.0142 • subsample: 0.9974 • colsample_bytree: 0.6391 • gamma: 0.6585 • reg_alpha: 0.9129 • reg_lambda: 0.6664
Support Vector Machine	• C: 0.003 • kernel: sigmoid • gamma: scale	• C: 1.18 • kernel: sigmoid • gamma: auto

Table 6 provides the performance metric values obtained for each model by addressing the multi-class and binary problems. Precision and recall were calculated by computing a weighted average of the number of actual observations associated with each class. Best values are highlighted in bold. For the multi-class approach, the XGBoost model obtained the lower value considering the cost metric provided by the challenge. However, for the other three metrics, the RF obtained the best values, with a lower difference for the cost metric with respect to the XGBoost. In contrast, for the binary approach, the RF obtained the best values for all the metrics considered. In addition, the lower value for the cost metric (36595) was obtained by addressing the problem as a binary problem, identifying this approach as the best solution for the problem under investigation. However, despite the encouraging results obtained for the cost metric, it is important to highlight that the overall accuracy remains below 75%. This suggests that, although the proposed model can potentially offer a significant reduction in the costs associated with the maintenance and/or replacement of Component X in SCANIA's heavy vehicles, its performance in terms of accurate fault identification over a long-time horizon is still limited. Consequently, its predictive reliability is not yet sufficient to fully support long-term preventive maintenance strategies.

The same challenge was solved by other authors by obtaining the performance prediction as listed in Table 7. This comparison demonstrated that our approach provided the best solution for the problem under investigation, with a lower cost metric than the best method previously reported in the literature known to us.

Finally, the prediction on the test set was obtained by adopting the RF trained. The confusion matrix shown in Table 8 was obtained, with a cost metric of 42976.

Beyond the specific context of the Scania dataset, the findings offer insights into predictive analytics for modern industry by showing how cost-sensitive evaluation metrics and automated hyperparameter optimization can strengthen

predictive maintenance frameworks. More broadly, the results highlight the importance of developing models that balance predictive accuracy with operational constraints, a requirement for effective deployment in complex industrial environments.

Table 6 Predictive performance of ML models.

Problem type	ML model	Cost metric	Accuracy (%)	Precision (%)	Recall (%)
Multi-class	Decision Tree	44756	49.66	95.7	49.66
	Random Forest	39722	**81.10**	**95.55**	**80.94**
	XGBoost	**37710**	66.69	95.8	66.68
	Support Vector Machine	59652	29.73	94.6	29.73
Binary	Decision Tree	36685	36.33	96.65	37.45
	Random Forest	**36595**	**72.16**	**95.83**	**72.94**
	XGBoost	37067	47.60	96.26	48.59
	Support Vector Machine	49566	1.50	0.04	2.7

Table 7 Comparison with other publications.

Authors	Reference	Approach adopted	Cost metric
Carpentier L. et al.	[17]	Survival analysis	50460
Parton M. et al.	[18]	Survival analysis	40109
Zhong J. and Wang Z.	[19]	Multi-class classification	41728
Our approach		Binary classification	36595

Table 8 Confusion matrix for the test set.

		Predicted label				
		0	1	2	3	4
True label	0	2906	0	0	0	1997
	1	7	0	0	0	19
	2	4	0	0	0	11
	3	20	0	0	0	21
	4	24	0	0	0	36

5. Conclusions

In this study, a structured approach for addressing the challenge of predicting failures of a component in truck engines manufactured by SCANIA AB in Sweden is presented. As a real-world dataset, it is characterized by missing values, anomalies, and imbalanced class distribution, typical challenges of industrial applications. To address these issues, we implemented a multi-step framework that systematically handled data preprocessing, feature selection and engineering, and model selection, ultimately leading to the construction of a robust predictive model. By reformulating the original multi-class problem, where failures were associated with different future time windows, into a binary classification task (failure vs. no failure), the proposed approach achieved superior performance in terms of the cost metric defined by the challenge, outperforming known approaches in the literature. This transformation was motivated by the cost structure specified in the challenge, which heavily penalized false positives, thus justifying a more conservative binary formulation.

Nonetheless, the proposed solution demonstrates a promising reduction in maintenance and replacement costs. However, it still exhibits limitations in accurately identifying failures, particularly in predicting the exact timing of the failure events. The overall accuracy remains below 75%, indicating potential enhancement, especially in applications where precise fault prediction is critical over extended operational periods. Future work will explore the adoption of survival analysis techniques to better capture the temporal dimension of component degradation and failure. The incorporation of time-to-event modelling has the potential to facilitate a more detailed comprehension of

the failure process, thereby supporting the development of predictive maintenance strategies that are both cost-effective and temporally precise. Furthermore, the adoption of deep learning models, which are capable of handling temporal sequences and uncertainty quantification, may enhance predictive reliability, particularly in cases involving long-term forecasts and rare failure events. Finally, the use of model interpretability techniques, such as feature attribution or hybrid explainable models, will be investigated to improve transparency and support informed decision-making in industrial predictive maintenance applications.

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References

[1] Shirole, V., Shahade, A.K., and Deshmukh, P.V. (2025) "A comprehensive review on data-driven driver behaviour scoring in vehicles: technologies, challenges and future directions." *Discov Artif Intell* **5 (26)**, doi:10.1007/s44163-025-00244-6.

[2] Hossain, Md N., Rahman, Md M., and Ramasamy, D. (2025) "Advances in intelligent vehicular health monitoring and fault diagnosis: Techniques, technologies, and future directions." *Measurement* **253, Part B**, doi: 10.1016/j.measurement.2025.117618.

[3] Ojeda, J.C.O., João de Moraes, J.G.B., Filho, C.V.d.S., Pereira, M.d.S., Pereira, J.V.d.Q., Dias, I.C.P., da Silva, E.C.M., Peixoto, M.G.M., and Gonçalves, M.C. (2025) "Application of a Predictive Model to Reduce Unplanned Downtime in Automotive Industry Production Processes: A Sustainability Perspective." *Sustainability* **17 (9)**: 3926, doi: 10.3390/su17093926.

[4] González-Prida, V., Márquez, C.P., Gunckel, P.V., Rodríguez, F.K., and Márquez, A.C. (2025) "Digital Transformation in Aftersales and Warranty Management: A Review of Advanced Technologies in I4.0." *Algorithms* **18 (4)**, doi:10.3390/a18040231.

[5] Adnan, I., Umer, T., Arsalan, A., Al Dabel, M.M., Bashir, A.K., and Ansif, A. (2025) "Data driven vehicular heterogeneity based intelligent collision avoidance system for Internet of Vehicles (IoV)." *Digital Communications and Networks,* doi:10.1016/j.dcan.2025.03.010.

[6] Schäfer, F., Gebauer, H., Gröger, C., Gassmann, O., and Wortmann, F. (2023) "Data-driven business and data privacy: Challenges and measures for product-based companies." *Business Horizons* **66 (4)**: 493-504, doi:10.1016/j.bushor.2022.10.002.

[7] Mahale, Y., Kolhar, S., and More, A.S. (2025) "Enhancing predictive maintenance in automotive industry: addressing class imbalance using advanced machine learning techniques." *Discov Appl Sci* **7**, **(340),** doi:10.1007/s42452-025-06827-3.

[8] Fayyad, U., Piatetsky-Shapiro, G., and Smyth, P. (1996) "From Data Mining to Knowledge Discovery in Databases". *AI Magazine* **17 (3)**: 37.

[9] Kharazian, Z., Lindgren, T., Magnússon, S., Steinert, O., and Reyna, O.A. (2025) "Scania component x dataset: A real-world multivariate time series dataset for predictive maintenance." *Sci Data* **12 (493)**, doi:10.1038/s41597-025-04802-6.

[10] Seabold, S., and Perktold, J. (2010) "Statsmodels: Econometric and Statistical Modeling with Python." *Proceedings of the 9th Python in Science Conference*: 92–96, doi: 10.25080/Majora-92bf1922-011.

[11] Chawla, N.V. , Bowyer, K.W., Hall, L.O., and Kegelmeyer. W.P. (2002) "SMOTE: synthetic minority over-sampling technique." *J. Artif. Int. Res*. **16 (1)**: 321–357, doi: 10.5555/1622407.1622416.

[12] Breiman, L. (2001) "Random Forests." *Machine Learning* **45**: 5–32, doi: 10.1023/A:1010933404324.

[13] Rokach, L., and Maimon, O. (2005). "Decision Trees. " In: Maimon, O., Rokach, L. (eds) Data Mining and Knowledge Discovery Handbook. *Springer*, Boston, MA, doi:10.1007/0-387-25465-X_9.

[14] Li, Z., Du, X., Xu, A., Wu, T., and Cao, Y. (2025) "Explaining tree ensembles through single decision trees " *Information Fusion* **123,** doi: 10.1016/j.inffus.2025.103244.

[15] Cervantes, J., Garcia-Lamont, F., Rodríguez-Mazahua, L., and Lopez, A. (2020) "A comprehensive survey on support vector machine classification: Applications, challenges and trends". *Neurocomputing* **408**, 189–215, doi: 10.1016/j.neucom.2019.10.118.

[16] Akiba, T., Sano, S., Yanase, O., Ohta, T., and Koyama, M. (2019). "Optuna: A Next-generation Hyperparameter Optimization Framework" arXiv, url: https://arxiv.org/abs/1907.10902.

[17] Carpentier, L., De Temmerman, A., and Verbeke, M. (2024). "Towards Contextual, Cost-Efficient Predictive Maintenance in Heavy-Duty Trucks". In: Miliou, I., Piatkowski, N., Papapetrou, P. (eds) Advances in Intelligent Data Analysis XXII. IDA 2024. Lecture Notes in Computer Science, vol 14642. Springer, Cham. doi:10.1007/978-3-031-58553-1_21.

[18] Parton, M., Fois, A., Vegliò, M., Metta, C., and Gregnanin, M. (2024). "Predicting the Failure of Component X in the Scania Dataset with Graph Neural Networks". In: Miliou, I., Piatkowski, N., Papapetrou, P. (eds) Advances in Intelligent Data Analysis XXII. IDA 2024. Lecture Notes in Computer Science, vol 14642. Springer, Cham. doi:10.1007/978-3-031-58553-1_20.

[19] Zhong, J., and Wang, Z. (2024). "Implementing Deep Learning Models for Imminent Component X Failures Prediction in Heavy-Duty Scania Trucks". In: Miliou, I., Piatkowski, N., Papapetrou, P. (eds) Advances in Intelligent Data Analysis XXII. IDA 2024. Lecture Notes in Computer Science, vol 14642. Springer, Cham. doi:10.1007/978-3-031-58553-1_22